

Elijah Tamarchenko

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Personal Details

Name Elijah Tamarchenko
Date of Birth 19.02.2001
Place of Birth Livingston, NJ, USA
Nationality USA

Education

Sept 2025 - Present **MSc Statistics**, ETH Zürich, Switzerland
Sept 2019 - June 2023 **B.A. Computer Science & Statistics with Highest Honors**, Williams College, USA
GPA: 3.85/4.0; Bachelors Thesis: *Combining Optimal Adjustment Set Selection and Post Selection Inference in Unknown Causal Graphs*
Spring 2022 **AIT Budapest, Budapest (Hungary)**
Exchange Semester; Final Grade: 5.0/5.0

Experience

July 2023 - June 2025 **Software Developer**, InterSystems Corporation
Developed and built open-source Git integration for InterSystems IRIS used by all clients. Parallelized AWS cloud deployments to improve scalability. Researched and developed RAG LLM architectures leveraging IRIS database capabilities. Applied machine learning methods and systems to model proprietary healthcare data
May 2022 - June 2023 **Machine Learning Researcher**, Williams College
Formulated novel theoretical statistical methodologies for estimating causal effects in hidden graph setting, presented research at American Causal Inference Conference. Developed estimation procedure to analyze counterfactual densities, improving accuracy over commonly used ATE
Sept 2020 - May 2023 **Teaching Assistant**, Williams College
Held Exercise sessions and lectures for the following courses:
○ STAT 442 Statistical Learning and Data Mining
○ STAT 346 Regression Theory and Applications
○ CSCI 256 Advanced Algorithm Analysis
○ CSCI 237 Computer Organization
○ CSCI 136 Data Structures and Algorithms
May 2021 - Sept 2021 **Machine Learning Engineer, Intern**, OneMarketData LLC
Applied machine-learning forecasting models to predict markets for business clients. Engineered data pipelining architecture for high-speed processing of Terabytes of telemetry market data

Awards and Honors

November 2025 **1st Place Prize Cycling Hack Zürich 2025**
Won Hackathon aimed to incentivize more people to cycle in Greater Zürich
February 2025 **1st Place Prize Intersystems Internal Hackathon 2025**
Won internal employee hackathon by developing a RAG-architecture for patient healthcare data
June 2023 **Sam Goldberg Colloquium Prize in Mathematics for Statistics**
Awarded for best colloquium talk in statistics at Williams College
Sept 2019 - June 2023 **V-Nee Yeh 1981 Scholarship**
Full scholarship to Williams College - 4 years
Oct 2019 **Questbridge Scholar**
Sept 2019 **National Merit Scholarship**
Awarded to top 1% of students nationally

(Selected) Skills

Programming Python, R, Javascript, SQL, Java, Go
Technologies Tensorflow, PyTorch, SkLearn, Docker, Angular, Slurm
Languages English (Native), Italian (Proficient), Russian (Proficient), German (A2)

(Selected) Projects

- 2023 **Undergraduate Honors Thesis**
Developed a parametric continuous optimization procedure that performs both covariate selection and effect estimation in a single step. The procedure often leads to a practical reduction in variance as shown in simulations
- 2022 **Research Abstract Classification**
Compared the performance of LSTM and transformers on classifying research papers based on their abstract, achieving an accuracy of 92% by finetuning models
- 2022 **Evolution Simulator**
A program that simulates natural selection and evolution through the reproduction of simple dot organisms over many generations, allowing analysis of various organism behaviors.
- 2021 **Analysis of School Funding on Student Achievement**
A causal analysis using machine learning techniques of school funding levels and student proficiency, which shows that decreases in school funding can account for about 18,000 students in MA being deemed not academically proficient.
- 2020 - 2021 **Williams Students Online**
Implemented professor review feature on the student website, used by over 2000+ students